

To: Dr. Alexis Smith, Editor, *The Open Journal of Astrophysics*
From: Sanskar Sontakke (Independent Researcher)
Date: September 2026
Subject: Resubmission of Revised Manuscript SGL-TDI-2026-R1

Dear Dr. Smith,

Please find enclosed the revised manuscript, “**Time-Domain Imaging of Rotating, Cloud-Covered Exoplanets with the Solar Gravitational Lens: Reconstruction Algorithms, Observing-Cadence Requirements, and Mission Targets**,” along with our point-by-point *Response to the Editor and Referees*.

We want to say plainly, before the details, that the review did more than tighten the presentation. Checking each quoted number against the shipped pipeline showed that several results in the submitted version could not have been produced by the code that produced the rest of them. We therefore re-ran the experiments rather than re-describing them, and the revised paper reports what the corrected pipeline actually measures. Some headline numbers moved, and one claim that we had made in the submitted version is now withdrawn outright. The main changes are these.

1. **The cadence law now has two explicit resource ledgers, and its numbers changed.** The referee was right that “8 h of photons per raster position” and “a 90-day campaign” cannot both be held fixed as the revisit count grows. We now run arm A (photons) and arm B (wall clock) separately and report every cadence figure per arm. Under arm A, r rises $0.049 \rightarrow 0.571$ from $M_p = 4$ to 64, monotonically and unanimously across six paired seeds; the arms agree within the paired standard error at every M_p , which is the evidence that the trend is a sampling effect and not a bookkeeping artefact. We also corrected the campaign length itself: with the 45 s per-dwell overhead ledger included, the fiducial campaign is **87.33 d** of wall clock, not the 90.0 d quoted previously, and the $M_p = 64$ arm-A configuration at 93.73 d is flagged as violating a 90-day cap rather than quietly presented inside one. The cloud-free control no longer looks uniformly excellent either ($r = 0.773$ at $M_p = 4$, rising to 0.987 at 64), so we report the conditioning contribution as real but front-loaded and saturating.
2. **Nuisance handling is now exact, and the attribution changed.** The submitted version’s “slot deflation” was an approximation that also rescaled the marginal noise. We replaced it with exact anchored profiling of the per-dwell-slot offsets, derived in closed form as a rank-one correction to the normal equations, and audited what it actually removes. The measured gain is $+0.054 \pm 0.006$ in r (10/10 paired seeds, $p_t = 1.1 \times 10^{-5}$) at 55% cover, replacing the previously quoted $+0.041$, $p < 10^{-3}$. Temporal whitening, by contrast, contributes nothing measurable at the fiducial cadence (-0.003 ± 0.002 , $p_t = 0.13$); we now say so rather than implying that it does.
3. **One submitted claim is withdrawn.** The submitted version stated that residual spin misalignments could be refined in flight by minimizing photometric residuals. They cannot. The whitened χ^2 is numerically invariant to the assumed initial phase ($\leq 10^{-11}$ relative) and monotone in the assumed period across the tested grid, so there is no minimum at truth to find. The spin state remains *observable* through reconstruction quality (r peaks at the true period for 4/4 seeds), and the ephemeris must come from external precursor astrometry. We measure the tolerance instead of assuming it: 5° of pole tilt costs 0.057 in r , 5° of initial phase costs 0.047, and 60° destroys the map.
4. **Optical validation is now described as what it is.** We adopted the published kernel and benchmark normalizations rather than recomputing them, and our discrete operator reproduces the analytic deconvolution-noise scaling to percent–tens-of-percent, not exactly (0.324 vs. 0.291 at $n = 64$; a $\sim 45\%$

gap at $n = 128$, which we state and attribute). The “machine precision” wording survives only where it is true — for the exposure-quadrature check, at 2.6×10^{-13} .

5. **Resolution and significance are banded, not point-estimated.** $r(\ell)$ is now computed with a real spherical-harmonic transform and Q_ℓ^m normalization: cloud-free it stays above 0.99 up to the $\ell = 20$ ceiling of the 36×72 grid, and under clouds it crosses 0.5 near $\ell \simeq 9$ (~ 2200 km). We quote that as an operating range and state that the crossing is not a resolution in the usual sense, replacing the submitted version’s single ℓ_{eff} value. For the land–ocean test we now run two nulls of different kind — a 40-member cloud-only *physical* null and a 2000-permutation label null — and report $p \leq 0.024$ as the ensemble floor rather than as a measurement, with $d' = 0.73 \pm 0.03$ (mean $\pm$ SE over 10 seeds) at 55.8% realized cover.
6. **The target table is a nominal-proxy ranking, not an uncertainty budget.** The submitted version presented propagated uncertainties that did not survive reproduction: the mass–radius dispersion quoted as 15% is the *Neptunian* branch of Chen & Kipping (the Terran branch is 4.03%), the radius was computed from $M \sin i$ while the text claimed an inclination correction that had not been applied, and the tracking budget used a scalar acceleration times window with an invented eccentricity term. The revised section gives the closed-form geometric de-inclination factors and labels them as conditional ratios rather than a posterior, reports the SNR_C brackets as bounded intervals over stated albedo and radius ranges with no confidence level attached, and replaces the tracking column with an explicitly defined along-window integral, $\Delta v = \oint |\mathbf{a}_\perp(t)| dt$, computed from Keplerian kinematics. τ Cet e is retained only as an illustrative dynamics row and labelled as such, because its planet interpretation has been refuted on activity grounds (Figueira et al. 2025, A&A 700 A174).
7. **Mission-level statements are now separated into adopted, derived, and out of scope.** Transportation, power and communications are adopted literature values with citations; the image-plane dynamics, dwell overheads and data volumes are derived here from our own kinematics. We do not size a propulsion system or select a departure site, and multi-craft swarms remain an asymptotic architectural scenario with registered visits per pixel separated from physical spacecraft count.
8. **Regularization is disclosed rather than defended.** The weight $\lambda = 3 \times 10^{-3}$ is a fixed constant in our implementation, not a fitted hyperparameter. A sweep over six decades shows the submitted version’s implicit insensitivity claim is wrong — cloudy r rises monotonically by $+0.178$ at $\lambda = 10$ while the objective stays flat — and since we have no truth-blind selection criterion to justify adopting the better-looking weight, we report this as a limitation instead of as a result.

Every number quoted above is reproduced from an archive shipped with the repository (`results/`), the experiment code is deterministic and seeded, and the change log in the response letter maps each referee comment to the file and section that now carries it. The revised manuscript is 24 pages with 8 figures and 13 tables.

We thank you and the referees for a review of unusual specificity. It changed the content of this paper materially, and in several places it changed our mind.

Sincerely,

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